

2.2.18

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Question: The unit vector normal to the plane $x + 2y + 3z - 6 = 0$ is:

Solution: The normal vector to the plane is

$$\mathbf{n} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad (0.1)$$

The magnitude of $\mathbf{n}$ is

$$\|\mathbf{n}\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}. \quad (0.2)$$

Therefore, the unit normal vectors are

$$\frac{\mathbf{n}}{\|\mathbf{n}\|} = \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad (0.3)$$

Plane and its Unit Normal Vector

